

Ans to the Q: NO: 1

Definition of Robot, specification and classification of Industrial Robot.

A robot is a programmable machine that can sense its environment, make decisions using a controller, and perform physical actions using motors or actuators. In simple words a robot can sense, think, and act to complete task automatically or semi-automatically.

For an example, a farming robot can detect dry soil, decide that watering is needed, and turn on water pump and it can also show the data to IoT and we can use Rule-Based AI for the watering system.

Importance Specification of an Industrial Robot :-

- ① Payload → The maximum weight that the robot can carry.
- ② Reach → The maximum distance the robot arm can reach.
- ③ Degree of freedom → Number of independent joints for movement.
- ④ Accuracy → How close the robot can reach the goal.
- ⑤ Speed → How fast the robot can move and done the work.
- ⑥ Controller Type → Electric, Hydraulic or pneumatic actuation.
- ⑦ Mounting Type → Floor mounted, ceiling mounted, wall mounted or mobile base.
- ⑧ Safety system → Emergency stop, sensors, and collision protection.

Industrial Robots basically used in factories and production lines.

Real examples: Universal Robots UR Cobots, FANUC Robotic Arms.

Importance Specification of a Social Robots :-

Social Robots are designed to interact with Humans. They can use speech, Facial expressions, gestures and AI based conversation.

- ① Communication ability → Can interact with voice, screen, gesture, or expressions.
- ② Sensors → Camera, microphone, touch Sensor, proximity and Lidar etc.
- ③ Actuators → Motors for movement of head, arms, wheel or Body.
- ④ AI Capability → Speech Recognition, Face Recognition, Emotion, Detection or conversation.
- ⑤ Connectivity → Wifi, Bluetooth, cloud, or mobile app connection.
- ⑥ Battery Life → How long the Robot can work after charging.
- ⑦ Safety → Safe movement around Humans.
- ⑧ Appearance → friendly, Human-like, animal like or toy-like Design.

Real example :- Pepper, NAO, Sophia, Furhat.

Classification of Industrial Robots :-

Type of Robot	Description	Example
Cartesian Robot	Moves in Straight X, Y and Z direction. It is accurate and Simple	3D printing, CNC machine, pick and place.
Cylindrical Robots	Has a Rotating base and moves vertically. It's working area cylindrical.	Machine loading, assembly work.
Spherical/ Polar Robot	uses rotary joints and one linear movement. It's work area is spherical	Welding and material handling
SCARA Robot	Fast in horizontal movement and good for small assembly tasks.	Electronics, painting, Packaging.
Articulated Robot	Looks like a Human arm with several rotary joints	Welding, painting, industrial assembly
Delta Robot	Has parallel arms and moves very Fast	Food sorting, High speed pick and place.
Collaborative Robot/Cobot	Designed to work safely with humans	Human-Robot teamwork in factories

Anst to the Q. No: 2Principle of sensors and their Uses in Robots:-① Proximity Sensor:

A proximity sensor detects nearby objects without touching them. It can work using infrared light, ultrasonic waves, magnetic fields, or capacitance. In Robots, it is used for obstacle detection, object counting, wall detection and safety.

example: A mobile Robot uses a proximity sensor to stop before hitting a wall.

② Gyroscope:

A gyroscope measures angular movement or rotation. It helps the robot understand whether it is tilting, turning or rotating. Robots use gyroscopes for balancing, direction control, drone stability and navigation.

example: A balancing robot uses a gyroscope to stay upright.

③ Tactile Sensor:

A tactile sensor detects touch, pressure, or force. It works like artificial skin for robots. It helps robots understand whether they are touching or holding something.

example: A robot gripper uses tactile sensors to hold an object without crushing it.

④ Infrared Sensor:

It uses infrared light. Usually, one part sends IR light and another part receives the reflected light. If an object is nearby, the reflected light changes.

example: A line-following robot uses IR sensors to detect black and white surfaces.

⑤ Lidar Sensors

LIDAR means Light Detection and Ranging. It sends laser light and measure how long the light takes to return after hitting an object. From this time, the robot calculates distance. that is used for mapping, obstacle avoidance, autonomous navigation and SLAM.

example: Autonomous vehicles use LiDAR to detect roads, objects and people.

⑥ Vision Sensors (Camera):

that is used for captures images or vedios. The Robot processes the image to identify object, faces, colors, QR codes, plants or paths.

Robot uses camera for object recognition, face detection, inspection, navigation, and human Robot interaction.

example: A farming robot can use a camera to identify plants or rows in a field.

⑦ Sound Sensors:-

A sound sensor or Microphone detects sound waves and converts them into electrical signals. Robots use sounds sensors for voice commands, elap control, speech Recognition, and sound-source Detection.

example: A social Robot use a microphone to understand Human voice command

Ans to the Q! NO! 3

Difference between Servo, Stepper, and DC motors :-

Feature	DC motor	Servo motor	Stepper motor
Basic function	Rotates Continuously When power is given	Moves to specific angle or position	Moves step by step in fixed angles
Control	Speed Controlled by voltage or PWM	position Controlled by PWM signals	position Controlled by pulses.
Feed back	Usually no built-in feedback	Has built-in feedback	usually works without feedback
Precision	Low to medium	Good position Control	very High position Control
Movement	Continuous Rotation	Usually 0° to 180° depending on type	Step by Step rotation
Cost	Low	Medium	medium to High
Common Use	Wheels, fans, pumps, small vehicles	Robot arms, steering Sensor rotation	CNC machine, 3D printers, accurate positioning.

Motors used in My Project :-

- ① DC geared motors for moving the 4-wheel robot.
- ② DC water pump motor for watering the plant.
- ③ Servo motor is not necessary in my update Design. because I am using three ultrasonic sensor to detect obstacle.

These are using in our project also to do it low cost and

Ans to the Q. NO. 4

Difference between Arduino UNO, Arduino Mega and Raspberry pi.

Feature	Arduino UNO	Arduino Mega	Raspberry pi
Type	Microcontroller Board	Microcontroller Board	Single-board Computer
Main Chip	ATmega328P	ATmega2560	Broadcom ARM processor
Operating System	No operating system	No operating system	Runs Linux/Raspberry pi OS
Digital pins	14 digital I/O pins	54 digital I/O pins	40-pin GPIO header.
Analog Pins	6 analog inputs	16 analog inputs	No built-in analog input.
PWM Pins	6 PWM pins	15 PWM pins	PWM possible through GPIO
Speed	16MHz	16MHz	Much faster, GHz-level processor
Memory	Small Memory	More memory than UNO	RAM in GB Range.
Programming	Arduino C/C++	Arduino C/C++	Python, C++, Linux tools
Best use	Simple Robotics and small projects	Large robotics projects with many sensor	AI, camera, web server, image processing.

Ans to the Q: NO: 5

Chart of the Social Robots with specifications, Market Cost and Bangladesh perspective.

Social Robot	Main purpose	Important Specification	Approx Market cost	Use in Bangladesh perspective
Miko 3	child education	AI learning, games, touch-screen, child friendly.	About \$299	Useful in schools, Coaching centers, Homes for interactive Learning.
Vector 2.0	Home Companion	Voice interaction, Camera, face recognition, small mobile body.	About \$199.99	Useful in Robotics labs and University for learning AI and HRI.
NAVO V6	Education and Research	Humanoid Robot, Cameras, microphone, Sensors, programming platform	About \$16,990	Useful in University for Robotics Research programming, and HRI experiments.
Pepper	Customer Service	Humanoid body, touchscreen, Camera, Sensors	Around \$16,990+	Useful in Banks, Hospital, airports, and shopping malls for guiding.
LOVOT	Emotional companion	Many sensor, emotional response, movement, touch interaction	\$5,000 - \$6,000	Useful in elderly care centers and Hospitals for emotional support
PARO	Therapy Robot	Seal-like Design, touch Response, sound Response, Therapy Support.	About £6,000	Useful in autism centers, mental health support.

Bangladesh Perspective :

In Bangladesh, Social Robot can be useful in education, Healthcare, elderly Care, Customer Services, and Robotics Research. However the main challenges are High cost, import dependency, maintenance, and lack of the Technical Support. If Low cost social Robots are developed Locally, they can become very helpful for schools hospitals, university and service sectors.

Ans to the Q: NO: 6

Project Title:- Knishi Rover: An Autonomus Farming Rover for Soil monitoring, Smart irrigation and Bangla voice Interaction.

Innovation of the Project:

The innovation of this project Rover is an Autonomus, Low Cost, and voice Assistance Futureistic project. We try to make it in low cost 4wd Rover and later on we will move it to a Shape which is with more feature. Based on the Review of some paper we saw most of are fixed position watering for whole Farming field. But our main goal and innovation is to build a mobile Robot that can move a small field and can show the temperature, Humidity, Soil condition and used Here Rule based AI and water only the specific area or spots. This Rover can the datas into IoT Dashboard, So the user can monitor Soil moisture and watering status.

The project Domain is Smart farming Agriculture Robot:

In this project the Rover detect the Farming Soil condition and check their either water need or Not. It alligns with Robotics cause the system uses sensor to collect data, a controller to make decisions, and motor pump to perform action.

So, we can say that the Rover work in the agriculture Field, but the technology uses Robotic movement, Sensor, IoT monitoring, automatic watering after the check soil to support smart farming.

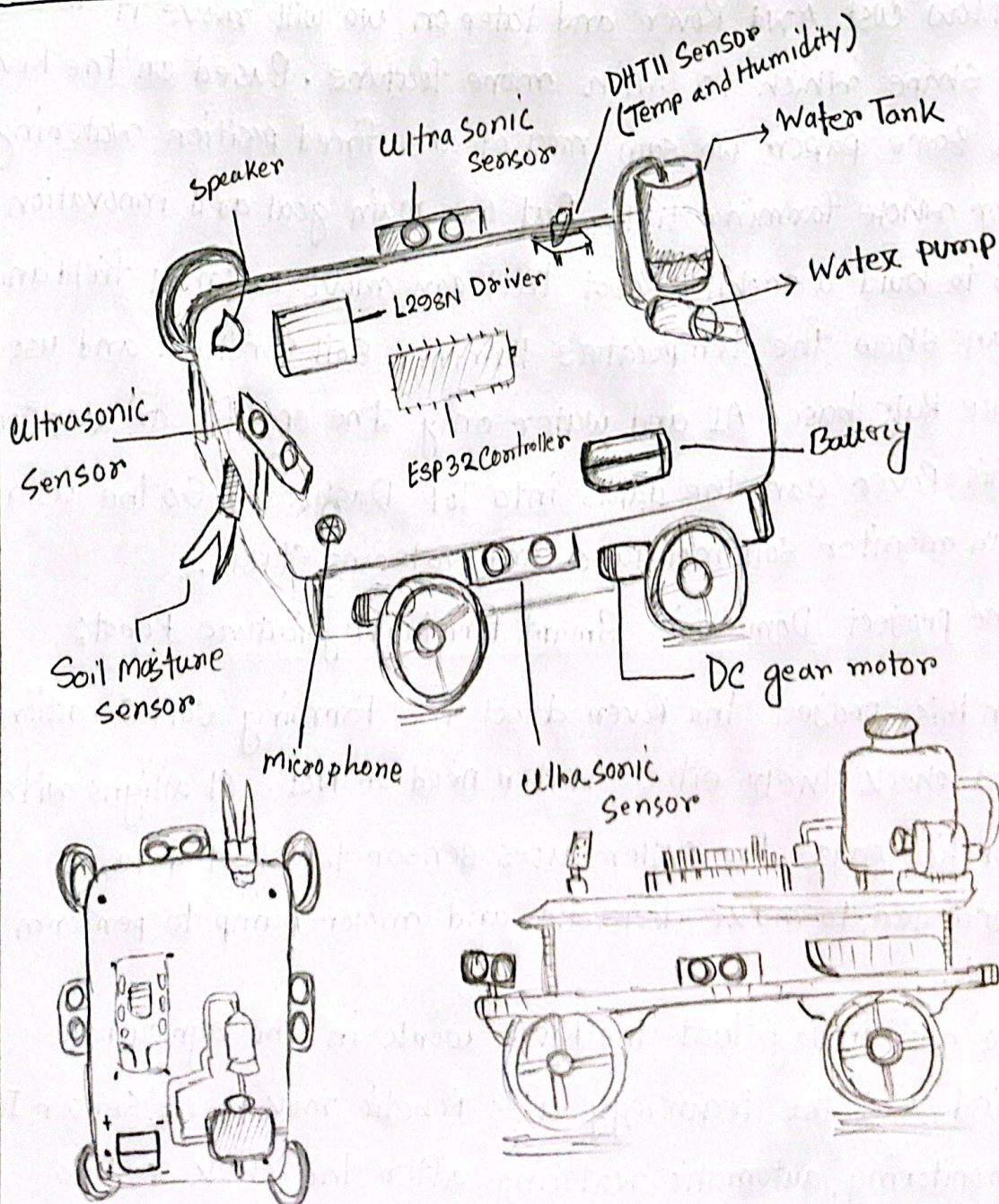
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Real world Application:

This Rover Can be used in agriculture Sections.

- ① Green House Farming
- ② Nursery plant Care.
- ③ Rooftop Gardening
- ④ Small agriculture field.
- ⑤ Small irrigation system.

Sketch the Rover for prototype:



Ans to the Q! NO: 7Human-Robot Interaction (HRI):

Human-Robot Interaction, or HRI is the study of how humans and Robots communicate, work together, and understand each other. It includes speech, gesture, safety, trust, design, usability and cooperation between human and Robots. Goodrich and Schultz describe HRI as a field focused on understanding, designing and evaluating robotic system for use ^{by} or with humans.

Example: A social Robot that talks with a patient in a hospital is an example of HRI

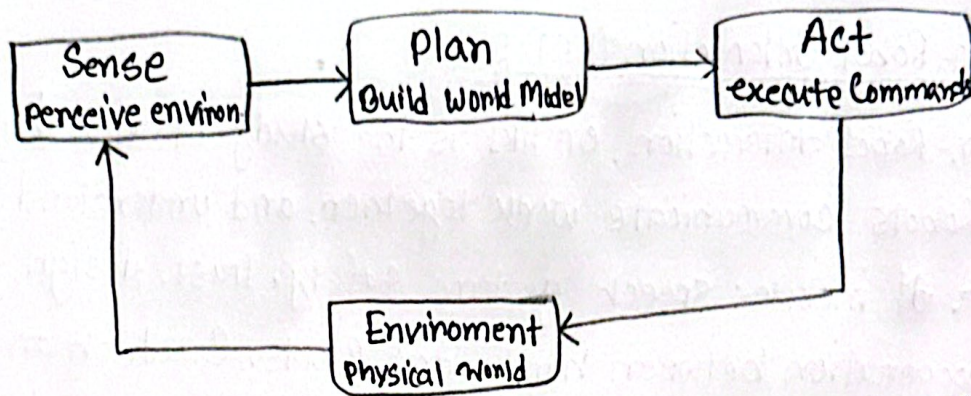
Asimov's Three Law of Robotics:

Isaac Asimov's Three Laws of Robotics are fictional but important ethical ideas in Robotics.

- ① A Robot must not harm a human being or allow a human to be harmed.
- ② A Robot must obey human orders, unless the order conflicts with the first law.
- ③ A Robot must perfect itself, unless this conflicts with the first or second law.

These laws were introduced in Asimov's Robot stories and are still discussed in Robotics ethics.

SPA model : Sense - Plan - Act



each stage runs once per cycle: Sense builds a model, plan reasons over the whole model to pick an action, act executes it, and the environment change is sensed again on the next loop. It's clean and predictable, but slow - every action waits on a full sense-plan cycle.

From project example:

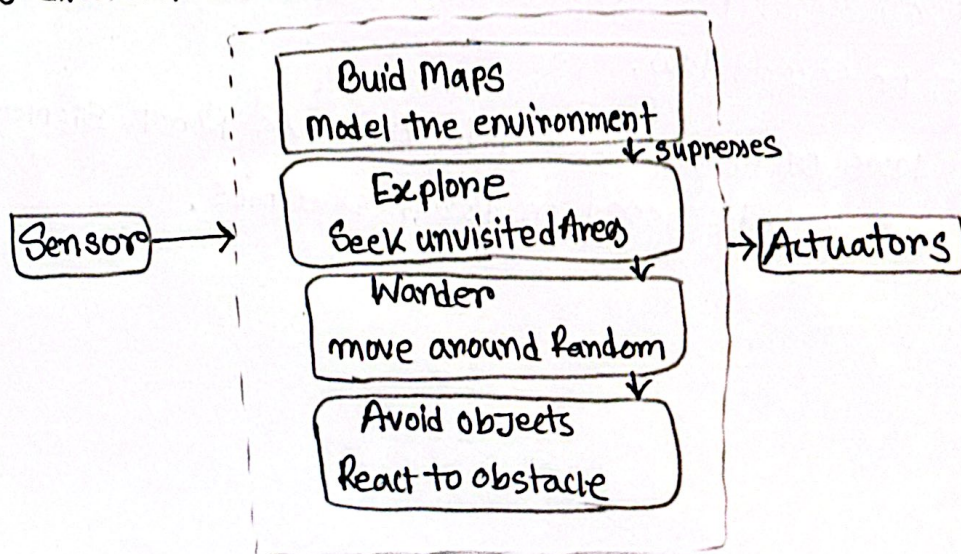
Sense : Soil Moisture Sensor detect dry zone.

Plan : ESP32 detect water needed.

Act : Water pump turns on and waters the plant.

Subsumption Behavior-Based Architecture:

That is a architecture is a Robot control method where Robot has different behavior layers.



Ans to the Q.NO.8

Anthropomorphism: that's mean giving human-like feature to non-human things. In Robotics, it means designing a Robot with human like appearance, voice, expression, movement or behavior.

example: A robot with eyes, smile, voice response and hand gesture is Anthropomorphism.

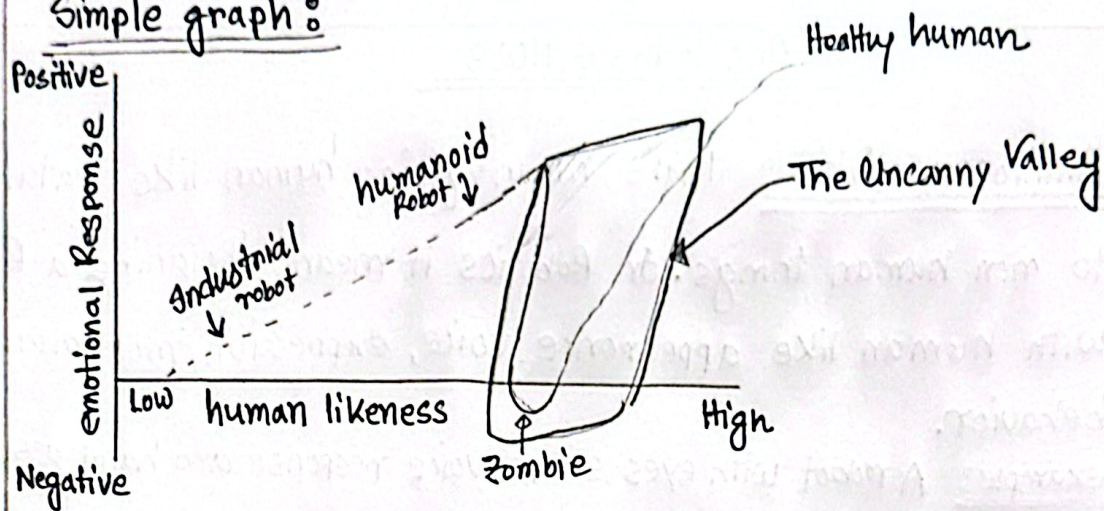
Uncanny Valley: that is a theory that explain human reaction to human-like Robots. When a Robot becomes more human like, people usually feel more comfortable with it. But if it becomes almost human but still look slightly unnatural, people may feel uncomfortable or scared.

This uncomfortable zone is called the Uncanny valley. Mori introduced the idea in 1970, and it's usually a graph human like-ness versus familiarity or comfort.

Theory: Mori theory Says -

- ① A machine like Robot maybe accepted cause people know it's a machine.
- ② A more human-like Robot may feel friendly and familiar.
- ③ But an almost human-like Robot with unnatural expression may feel Creepy.
- ④ A Real Human again gives the highest comfort and Familiarity.

Simple graph :



A good Robot design follow these steps :

- ① Identify the problem : What kind of problem Robot Solve.
- ② Understand the users : Know who will use the Robot and where it will work
- ③ Collect Requirements : Sensors, motor, power supply etc.
- ④ Create concept Design : Draw sketches, block Diagrams, and system flow.
- ⑤ Build prototype : first working model.
- ⑥ Test the Robot : Test sensor, motor, power etc.
- ⑦ Take feedback : Ask users, Faculty, Supervisor.
- ⑧ Improve the design : Fix problems and Look.
- ⑨ Final Implementation : Prepare final Demo, Report and presentation.

Ans to the Q. NO. 9User Centered Design :

that is an approach where the robot is designed according to the needs, comfort, ability, and limitations of the user.

In this method designers study the users, understand their problems, build prototype and test the system.

Participatory Design:-

that is a Design approach where user directly involved in the design process. Users do not only test the robot, they also give ideas, suggest features and help make design decisions.

Differentiation between "User Centered" and "Participatory"

Topic	User Centered	Participatory
User Role	User is studied and observed	User works as a Co Designer
Decision Making	Mainly done by designers	Shared between designers and users
Feedback	usually taken after prototype	Taken from the beginning.
Main focus	Usability and user mode	Cooperation and user involvement.
Example	Farmer test the Robot after Development	Farmer Help to Design the Robot from the beginning.

Ans to the Q: NO. 10Types of Nonverbal Interaction :-

Type	explanation	Robot example
Gesture	Movement of hand, arm, or body to express meaning.	Robot waves hand to say Hello.
facial expression	Showing emotion through face or Display.	Robot shows happy or sad on screen.
Gaze/ EyeContact	Looking direction shows attention	Robot looks at the person who is talking.
Posture	Body positioning express or emotion	Robot leans forward to show interest.
Proxemics	Maintaining proper distance from humans	Robots stop at a safe distance.
Touch/Haptics	Communication through touch	Robot reacts when user touches.
Body movement	movement style shows behavior	Robot moves slowly near human for safety.
Appearance	Shape and design communicate purpose.	Cute design for children, serious design for industry.

Nonverbal interaction is important because human can understand the robot more naturally without long explanations.